

Country-Pack Staged Assurance and Development Decoupling Prompt

Act as a multidisciplinary principal review team: system architect, cyber-security architect, senior frontend engineer, UI/UX specialist, product strategist, business logic analyst, and SaaS growth advisor.

Act as a senior enterprise country-pack and regulatory-isolation architecture team:

- Enterprise software architect: separate core product development from jurisdiction-specific regulatory certification.
- Payroll and compliance architect: distinguish development assumptions from legally approved production rules.
- Release-assurance specialist: place controls at the correct promotion boundary instead of blocking ordinary development.
- Cybersecurity and RBAC specialist: preserve tenant isolation, auditability, maker–checker controls, and production safeguards.
- Developer-experience specialist: minimize unnecessary approvals, manual intervention, and integration friction.
- OHADA/SYSCOHADA-aware platform architect: keep statutory configuration effective-dated, replaceable, and independently reviewable.
- SaaS modularity specialist: ensure an unfinished country pack cannot prevent unrelated platform capabilities from being developed, integrated, or tested.

Objective

Assess why independent expert approval and signed regulatory evidence are currently required for country-pack development, and determine whether these controls can be moved from the development workflow to the jurisdiction-validation and production-promotion boundary.

The preferred outcome is a clean, enterprise-grade model in which:

- core features can be developed and integrated without regulator signatures;
- country-pack-dependent features can operate in development and sandbox using clearly labelled provisional fixtures;
- automated tests can use synthetic or unapproved country-pack data;
- missing legal approval does not block unrelated builds, tests, merges, deployments, or feature integration;
- regulatory documentation, independent review, signatures, and production authorization become mandatory only when a specific country pack is promoted for live statutory use.

Do not simply remove the controls. Determine which controls are necessary, when they are necessary, and how to relocate them to the narrowest legitimate boundary.

Questions to answer

1. Why do these approvals exist?

Explain precisely what risks are controlled by:

- `source_artifact_hash_verification`
- `source_artifact_expert_approval`
- independent fixture tie-outs;
- reviewer qualifications and conflict declarations;
- signed approval artifacts;
- maker–checker authorization;
- `REGULATOR_CONFIRMED` OR `EXPERT_REVIEWED` classifications.

Distinguish protection against:

- incorrect payroll calculations;
- obsolete rates or ceilings;
- unsupported legal interpretations;
- false product claims;
- unauthorized production use;
- audit and liability exposure;
- evidence tampering.

Identify which risks apply during ordinary software development and which apply only when a country pack is used in production.

2. Are the controls currently positioned correctly?

Inspect whether regulatory approval requirements are unnecessarily coupled to:

- application builds;
- feature development;
- unit and integration tests;
- database migrations;
- UI development;
- unrelated modules;
- CI pipelines;
- development deployments;
- sandbox demonstrations;
- production releases that do not enable the affected country pack.

Identify every point where an unapproved country pack creates avoidable development friction.

Use `graphify-out/` and the relevant source files to determine the dependency and gate relationships.

3. Design a staged assurance model

Evaluate and, if justified, implement the following lifecycle:

Stage A — Core development

Permitted without regulatory signatures:

- core platform development;
- generic payroll and accounting architecture;
- country-pack interfaces and schemas;
- feature integration;
- unit, integration, contract, and UI testing;
- synthetic fixtures;
- non-jurisdictional workflows.

Required safeguards:

- no live statutory claims;
- no production country-pack activation;
- no live declarations or authority submissions;
- no use of provisional results as legally approved payroll output.

Stage B — Country-pack development

Permitted with official sources but without completed expert approval:

- provisional country-pack configuration;
- development-only calculations;
- golden-fixture development;
- internal demos;
- sandbox testing;
- integration and regression testing.

Required status:

- DRAFT, SOURCE_VERIFIED, OR PENDING_EXPERT_REVIEW;
- productionUseAllowed: false;
- visible provisional-result warnings;
- fail-closed authority and production adapters.

Stage C — Jurisdiction validation

Required before production promotion:

- official source verification;
- effective-date review;
- independent calculations;
- complete fixture-family tie-outs;
- qualified reviewer identity and competence;
- conflict declaration;
- signed approval artifact;
- maker–checker verification;
- explicit production-use decision.

Stage D — Production activation

Required controls:

- country-pack-specific production gate;
- tenant and jurisdiction entitlement;
- effective-date resolution;
- approved artifact hashes;
- rollback and revocation support;
- audit evidence;
- monitoring for expiry or superseding rules.

Determine whether this model satisfies the repository's risk and release requirements.

4. Separate platform release from country-pack promotion

Design independent release tracks:

1. platform release;
2. country-pack development publication;
3. country-pack production certification;
4. tenant-specific country-pack activation.

An unapproved Cameroon pack should block only:

- live Cameroon statutory calculations;
- Cameroon production declarations;
- production authority submissions;
- claims that the Cameroon implementation is legally approved.

It should not block unrelated product development or deployment.

Define the exact state machine, for example:

DRAFT → SOURCE_VERIFIED → READY_FOR_EXPERT_REVIEW → EXPERT_REVIEWED → PRODUCTION_ELIGIBLE → ACTIVE → SUSPENDED/REVOKED

Specify who may perform each transition and which transitions require human approval.

5. Establish provisional-mode behavior

Define how provisional country packs operate safely:

- explicit environment restrictions;
- synthetic or masked data only where necessary;
- prominent provisional labels;
- no production adapters;
- no live payments or declarations;
- deterministic test fixtures;
- audit logs;
- expiration and review dates;
- safe fallback when a pack is missing or unapproved.

Determine whether provisional country-pack results may be displayed to developers or test users and how they must be labelled.

6. Minimize human approval

Identify every approval currently required and classify it as:

- legally or operationally essential;
- required only at production promotion;
- automatable;
- replaceable with deterministic evidence;

- redundant and removable.

Automate where safe:

- source downloads and provenance capture;
- SHA-256 computation;
- schema validation;
- fixture comparison;
- effective-window checks;
- evidence completeness;
- signature-file integrity;
- gate reporting;
- expiry and supersession alerts.

Keep human review only for professional interpretation, approval, independence, and production authorization.

7. Prevent end-stage surprises

Do not postpone all evidence work until the final day.

Define a lightweight continuous-evidence process during development:

- retain official sources when a fixture is introduced;
- record provisional legal references;
- maintain fixture traceability;
- identify unresolved interpretations;
- generate the reviewer packet automatically;
- keep signatures and final approval deferred until production promotion.

The final review should validate an already organized evidence package, not reconstruct months of undocumented decisions.

Evidence to inspect

Inspect:

- docs/country-pack/COUNTRY_PACK_REGULATORY_ISOLATION_TARGET_ARCHITECTURE_2026-07-26.md
- docs/country-pack/COUNTRY_PACK_REGULATORY_ISOLATION_IMPLEMENTATION_2026-07-26.md
- docs/country-pack/CAMEROON_CNPS_EXTERNAL_REVIEW_AND_SIGNATURE_ENABLEMENT_IMPLEMENTATION_2026-07-26.md
- docs/HR-Payroll/evidence/country-packs/CM/2026-07-19/
- services/regulatory/country-packs/
- services/regulatory/runtime/
- services/regulatory/ports/
- relevant compliance and payroll services;
- statutory development, review-preflight, production, policy, and enterprise release gates;
- relevant tests and CI commands;
- graphify-out/ dependency reports.

Preserve the dirty worktree and do not modify unrelated files or lint findings.

Required output

Produce:

1. an evidence-based explanation of why the approvals exist;
2. a determination of which approvals are unnecessary during development;
3. a dependency and gate-impact analysis;
4. a staged country-pack assurance architecture;
5. a state-transition and authorization model;
6. a platform-release versus country-pack-promotion matrix;
7. a list of gates to retain, relocate, split, automate, or remove;
8. a provisional development-mode specification;
9. a migration plan that does not weaken production protection;
10. focused implementation changes where justified;
11. tests proving development remains unblocked while production fails closed;
12. residual risks and external actions;
13. a final recommendation.

Save the report under:

`docs/country-pack/`

Save the implementation follow-up under:

`what-next/`

Verification

At minimum, verify:

```
npm run statutory:country-pack:dev:gate
node scripts/statutory-country-pack-review-preflight.js --mode report
node scripts/statutory-country-pack-production-gate.js --mode report
```